Research Topics in Blockchain
&
Blockchain and Its Application



Lecture # 12:
Why this Consensus mechanism Works

- ✓ **Incentive and Proof of work**
- ✓ **Putting it all to geather**

Nodes in the Bitcoin network do not have real-world identities, making punishment difficult.

•**Solution: Incentivizing Nodes with Bitcoin:**
•Instead of punishing dishonest nodes, Bitcoin rewards hone
•Since Bitcoin itself is a digital currency, it is used as a rewar

•**Block Reward Mechanism:**

•The miner who successfully creates a block gets a reward t
coin-creation transaction.

•**Bitcoin Halving Process:**
•Initially, the block reward was **50 BTC**, but it halves every **2**
years).
•The reward was **25 BTC** at the time of writing and continues
•This ensures a finite total supply of **21 million bitcoins**.
•The reward is assigned to an address chosen by the miner, usually their own.

•When a miner successfully mines (solves the cryptographic puzzle) and adds a new block to the blockchain, they receive a reward.
•This reward comes in the form of newly created bitcoins (or the native cryptocurrency of the blockchain) and is known as the **block reward**.
•The reward is given through a **special transaction** called the **coinbase transaction**, which is the first transaction in every block.
•The miner includes their own address in this transaction, allowing them to claim the newly minted coins.

Let's assume the following scenario where **Alice, Bob, and a Miner** are involved in transactions:

Step 1: Initial Balances (Before Any Transactions)

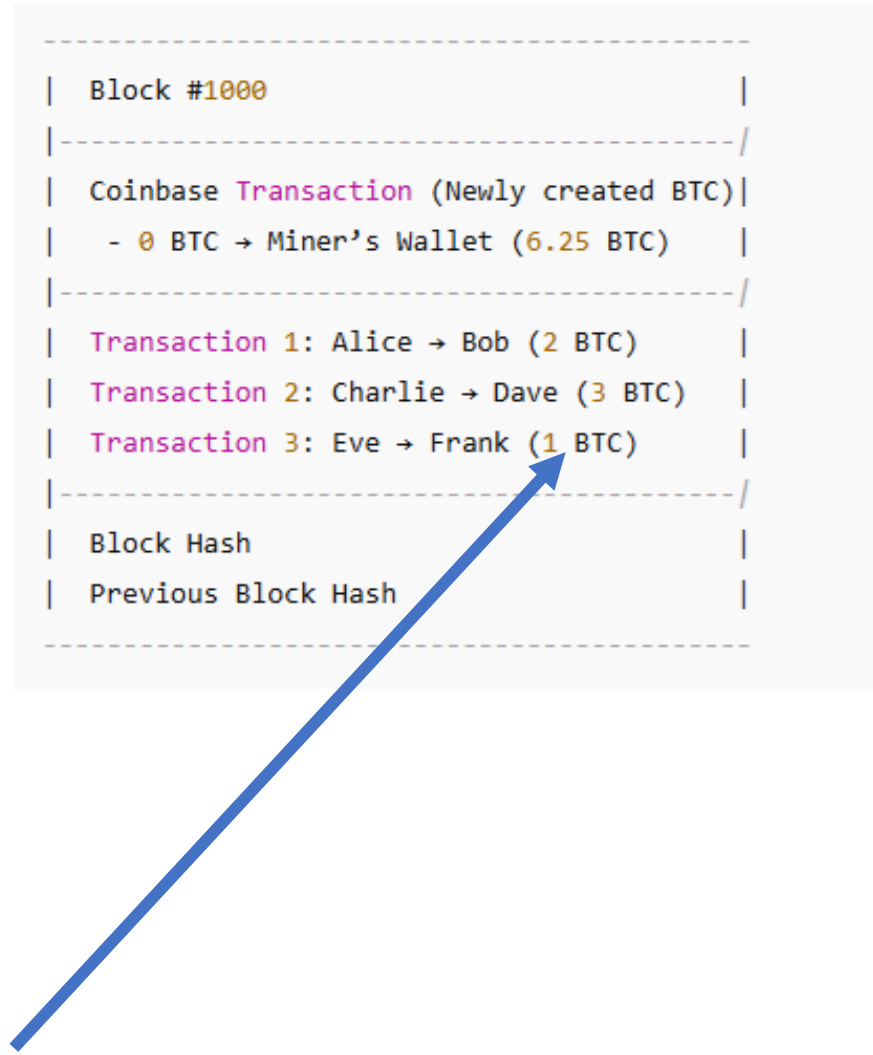
Address (Wallet)	Balance (BTC)
Alice	10 BTC
Bob	5 BTC
Miner	0 BTC

Step 2: Alice Sends 2 BTC to Bob

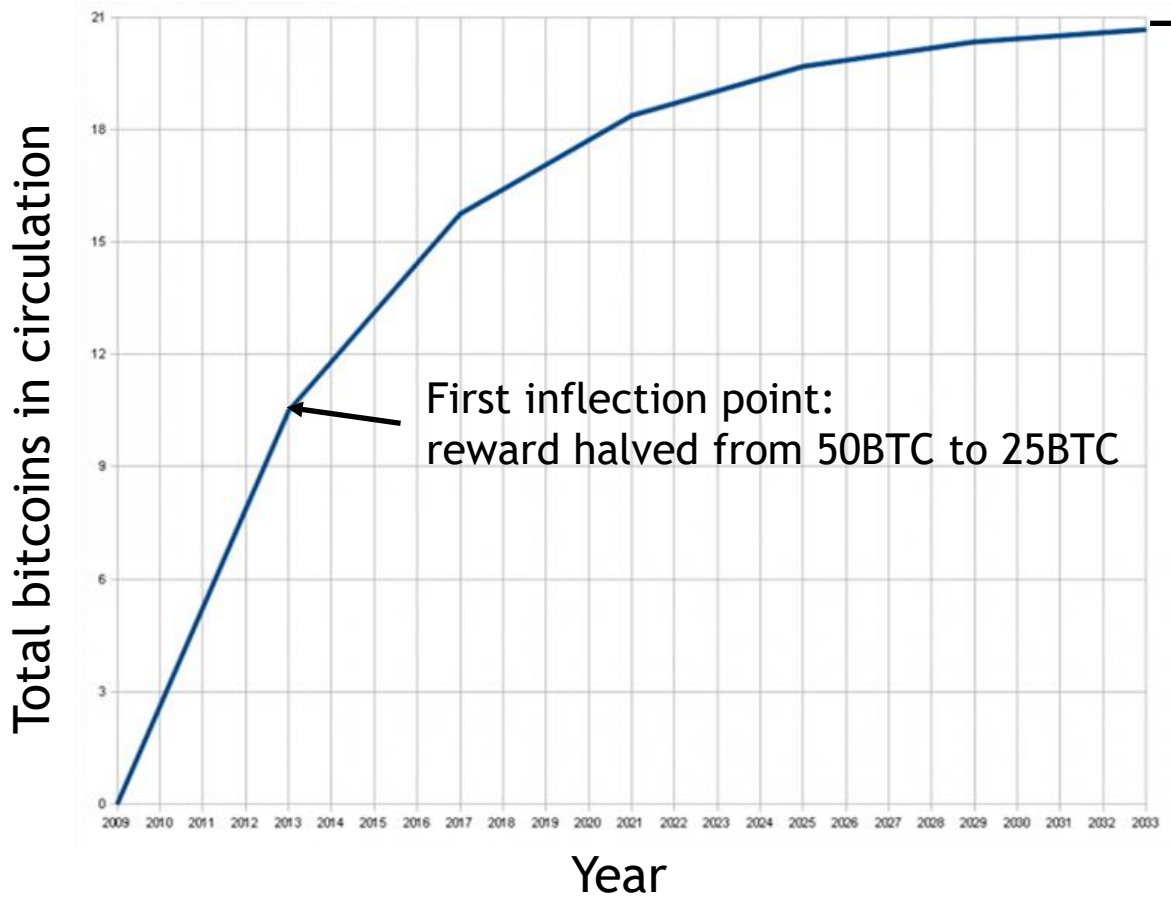
- Alice creates a transaction:
  Alice → Bob (2 BTC)
- This transaction is broadcast to the network.
- Miners compete to add this transaction into a block.

Step 3: The Miner Successfully Mines the Block

- The miner solves the cryptographic puzzle and creates a new block.
- The block contains:
  - Alice's transaction (**Alice → Bob, 2 BTC**).
  - A **coinbase transaction**, which rewards the miner with **6.25 BTC** (current Bitcoin block reward).



There’s a finite supply of bitcoins



Total supply: 21 million

Block reward is how new bitcoins are created

Runs out in 2040. No new bitcoins unless rules change

Incentive 2: transaction fees

What is a Transaction Fee in Bitcoin?

- A **transaction fee** is an additional amount that users pay to miners when they send Bitcoin (or any cryptocurrency). This fee incentivizes miners to prioritize and confirm transactions faster.

Why Do Transaction Fees Exist?

1. **Compensation for Miners** – Miners receive transaction fees as a reward for validating and adding transactions to the blockchain.
2. **Faster Confirmation** – Transactions with **higher fees** are processed more quickly since miners prioritize them.
3. **Security & Spam Prevention** – Fees prevent users from flooding the network with unnecessary transactions

Example of a Bitcoin Transaction with Fees:

Imagine **Alice** wants to send **1 Bitcoin** to **Bob**.

•The total amount she has in her Bitcoin wallet is **1.001 BTC**.

•She decides to send **1 BTC** to Bob and **0.001 BTC** as a transaction fee.

Transaction Breakdown:

- **Sender:** Alice
- **Receiver:** Bob
- **Total Sent: 1 BTC**
- **Transaction Fee: 0.001 BTC**
- **Final Amount in Bob's Wallet: 1 BTC**
- **Miners Receive: 0.001 BTC as a reward**

Remaining problems

1. How to pick a random node?

1. How to avoid a free-for-all due to rewards?

1. How to prevent Sybil attacks?

Proof of work

To approximate selecting a random node:
select nodes in proportion to a resource
that no one can monopolize (we hope)

- In proportion to computing power: proof-of-work
- In proportion to ownership: proof-of-stake

Equivalent views of proof of work

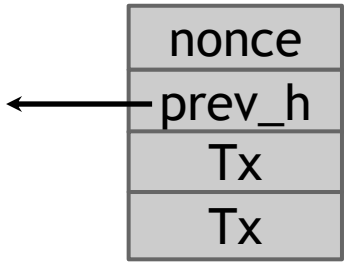
1. Select nodes in proportion to computing power

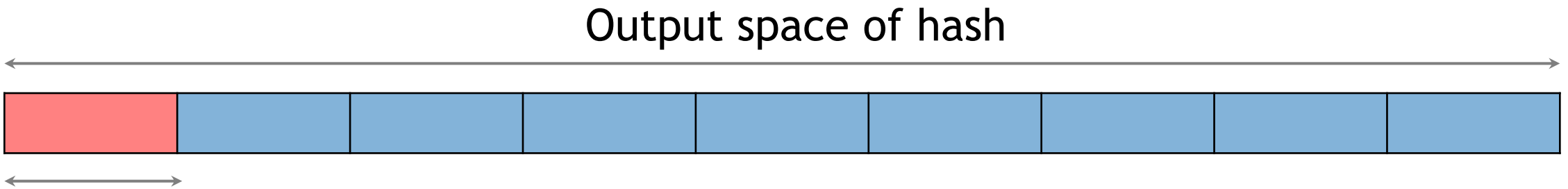
1. Let nodes compete for right to create block

1. Make it moderately hard to create new identities

Hash puzzles

To create block, find nonce s.t.
H(nonce ‖ prev_hash ‖ tx ‖ … ‖ tx) is very small





If hash function is secure:
only way to succeed is to try enough nonces until you get lucky

PoW property 1: difficult to compute

As of Aug 2014: about 10^{20} hashes/block

Only some nodes bother to compete — miners

PoW property 2: parameterizable cost

Nodes automatically re-calculate the target every two weeks

Goal: average time between blocks = 10 minutes

Prob (Alice wins next block) =
fraction of global hash power she controls

Every 2016 blocks (~two weeks), the Bitcoin network automatically recalculates this target to maintain an average block time of **10 minutes**.

- If blocks are being mined too quickly (less than 10 minutes on average), the difficulty **increases**.
- If blocks are being mined too slowly (more than 10 minutes on average), the difficulty **decreases**.

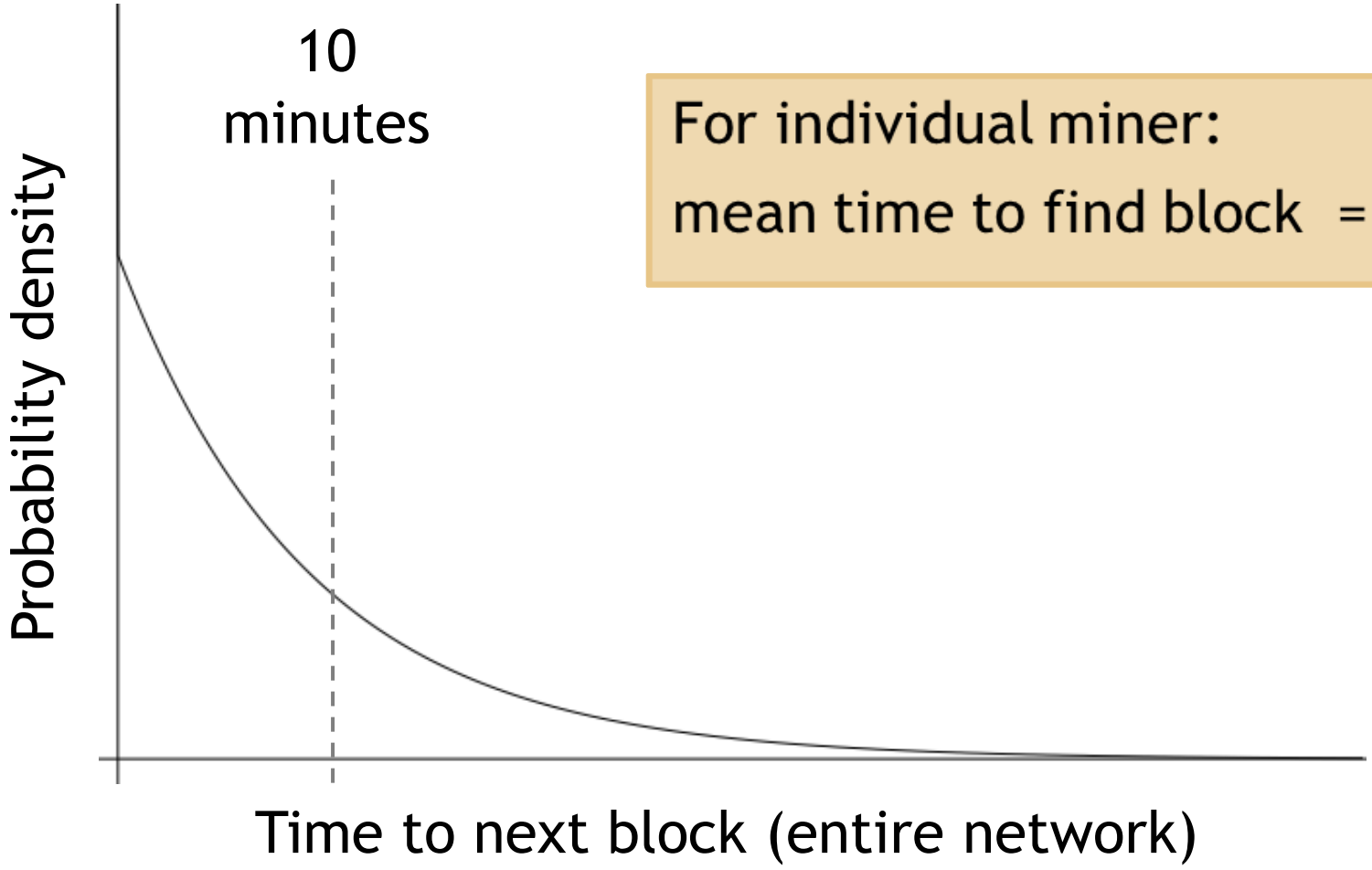
Why is this important?

- It keeps the Bitcoin network stable and predictable, **preventing blocks from being mined too fast or too slow**.
- It ensures fairness and prevents a sudden increase in mining

Key security assumption

Attacks infeasible if majority of miners weighted by hash power follow the protocol

Solving hash puzzles is probabilistic



For individual miner:

$$\text{mean time to find block} = \frac{10 \text{ minutes}}{\text{fraction of hash power}}$$

PoW property 3: trivial to verify

Nonce must be published as part of block

Other miners simply verify that
$H(\text{nonce} \| \text{prev_hash} \| \text{tx} \| \ldots \| \text{tx}) < \text{target}$

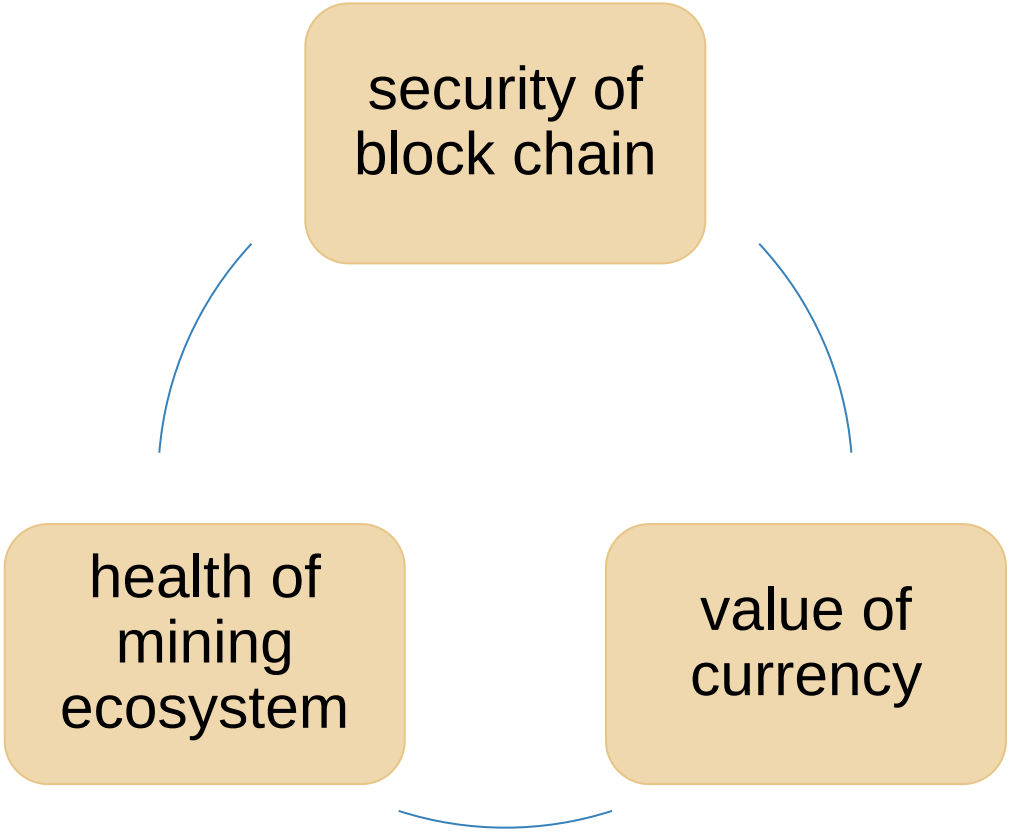
Mining economics

If mining reward (block reward + Tx fees)	>	hardware + electricity cost	→	Profit

Complications:

- fixed vs. variable costs
- reward depends on global hash rate

Bitcoin is bootstrapped



What can a “51% attacker” do?

Steal coins from existing address? ✗

Suppress some transactions?

- From the block chain ✓
- From the P2P network ✗

Change the block reward? ✗

Destroy confidence in Bitcoin? ✓✓

Remaining questions

How do we get from consensus to currency?

What else can we do with consensus?